

# Classical Field Theory

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# Preface

This is my notes on Classical Field Theory. The main reference is *Classical Electrodynamics* by John David Jackson. Other references include 电动力学 by 蔡圣善 and 电动力学 by 郭硕鸿.

# Chapter 1

## Introduction

经典物理中，有两个东西需要用场来描述。

- 多体系统：引入密度分布  $\rho(\vec{r}, t)$  和速度场  $\vec{v}(\vec{r}, t)$  来描述流体
- 相互作用：电动力学中用电场  $\vec{E}(\vec{r}, t)$  和磁场  $\vec{B}(\vec{r}, t)$  来描述电磁相互作用

# Chapter 2

## Classical Field Theory

### 2.1 Classical Scalar Field

**Definition 2.1.1.** a smooth map  $\phi : M \times \mathbb{R} \rightarrow \mathbb{R}/\mathbb{C}$  is a real/complex scalar field on manifold  $M$ .

*Remark 2.1.2.* Scalar field is usually written as  $\phi(x, t)$ , and  $x \in M$ ,  $t \in \mathbb{R}$  represent time. In SR and GR, we usually treat space and time equally, but in classical mechanics we usually divide space and time.

### 2.2 Representation of Galilean Group

### 2.3 Classical Spinor Field

### 2.4 Classical Vector Field

# Chapter 3

## Relativistic Field Theory

### 3.1 Relativistic Scalar Field

### 3.2 Representation of Lorentz Group

### 3.3 Relativistic Spinor Field

### 3.4 Relativistic Vector Field

## Chapter 4

# Lagrangian Formalism of Classical Field Theory

## Chapter 5

# Hamiltonian Formalism of Classical Field Theory

# Chapter 6

## Symmetry

对于一个映射，只要这个映射的作用是把一个场映射成一个新的场，那么就把这个映射称为一个变换。

自然可以用流形上的微分同胚来构造场的变换。

Consider a transformation  $\phi(x) \rightarrow \phi(f(x))$ ,  $x \rightarrow x$ , and  $f : M \rightarrow M$  is a diffeomorphism.

**Theorem 6.0.1.** *Let  $\phi$  be a field on smooth manifold  $M$ . The iso of  $M$  is  $Iso(M)$  and the symmetry group of  $\phi$  is  $Sym(\phi)$ . Then  $Iso(M)$  is a subgroup of  $Sym(\phi)$ .*

**证明.** Let  $g$  be the metric of  $M$  and  $S[\phi, g]$  be the action of  $\phi$ . Let  $\varphi$  be a diffeomorphism  $\varphi : M \rightarrow M$ .

The action of field is invariant under any coordinates transition, which means

$$S[\varphi^* \phi, \varphi^* g] = S[\phi, g] \quad (6.1)$$

If  $\varphi \in Iso(M)$ , then

$$\varphi^* g = g \quad (6.2)$$

So

$$S[\varphi^* \phi, g] = S[\phi, g] \quad (6.3)$$

which means that  $\varphi \in Sym(\phi)$ . Hence  $Iso(M)$  is a subgroup of  $Sym(\phi)$ .  $\square$

**Example 6.0.2.** Considering a scalar field  $\phi$  on a Minkowski manifold  $M$ , The Lagrange density of  $\phi$  is

$$L = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m \phi^2 \quad (6.4)$$

The corresponding action is obviously do not change in any coordinates tansfoemation.

Lorentz transformation is a transformation defined below.

**Definition 6.0.3.** A transformation

# Chapter 7

## Fluid Mechanics

## Chapter 8

### Time-independent Electronic Field

## Chapter 9

# Time-independent Magnetic Field

# Chapter 10

## Maxwell's Equations

# Chapter 11

## Electromagnetic Wave

### 11.1 Electromagnetic Wave in Vacuum

### 11.2 Diffraction

# Chapter 12

## Quasi-Static Field

# Chapter 13

## Waveguide

## Chapter 14

# Electromagnetic Radiation